

That being said, having good testing to training ratio is preferred with most simulations using a ten cross-fold validation. Another aspect happens to be the issue with biases and variances. Biases represent how far the values differ from the target values, and variances relate to the difference between the values themselves.

Any good machine learning program will try to minimise both and in most cases develop a trade-off between the two, to correctly identify each data instance.

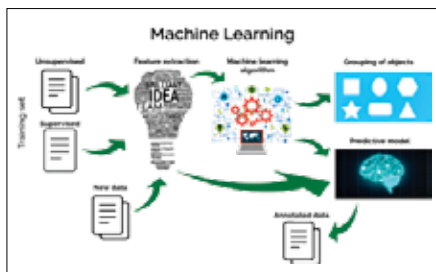
Tracking machine learning in real life

Apart from the more lucrative cases of machine learning being used in self-driving cars and personalised assistance, which we'll discuss in the next segment, let's stroll past some simple cases right from the history of data:-

Google's Flu Predictor: Having begun on the project of collecting search data for long periods of time, Google published findings of the search dynamics for flu-related cures all over the world. The algorithm was supported by a series of support vector machines (SVM) that measured search indexes on a number of parameters.

Based on past data, the company was able to distinguish patterns throughout the year and concentrate on regions that reported the highest of such searches. This enabled the company to build a theoretical model that predicted values for stock flu searches in the coming months and

thus alert health officials to stockpile on medicines in regions with the highest number of searches.



Typical Machine Learning Workflow

Ohio University's CHD

Case: Here's a fun little machine learning exercise that uses a well-known technique called logistic regression. Analysts from the University of Ohio had

spent many years in studying the incidence of coronary heart disease(CHD) in patients classified by age groups, gender, race, lifestyle and dietary habits.

Logistic regression is a special type of machine learning technique where the output variable can take two values. In this case, an output of 1 meant that the patient would be diagnosed with CHD and 0 meant that